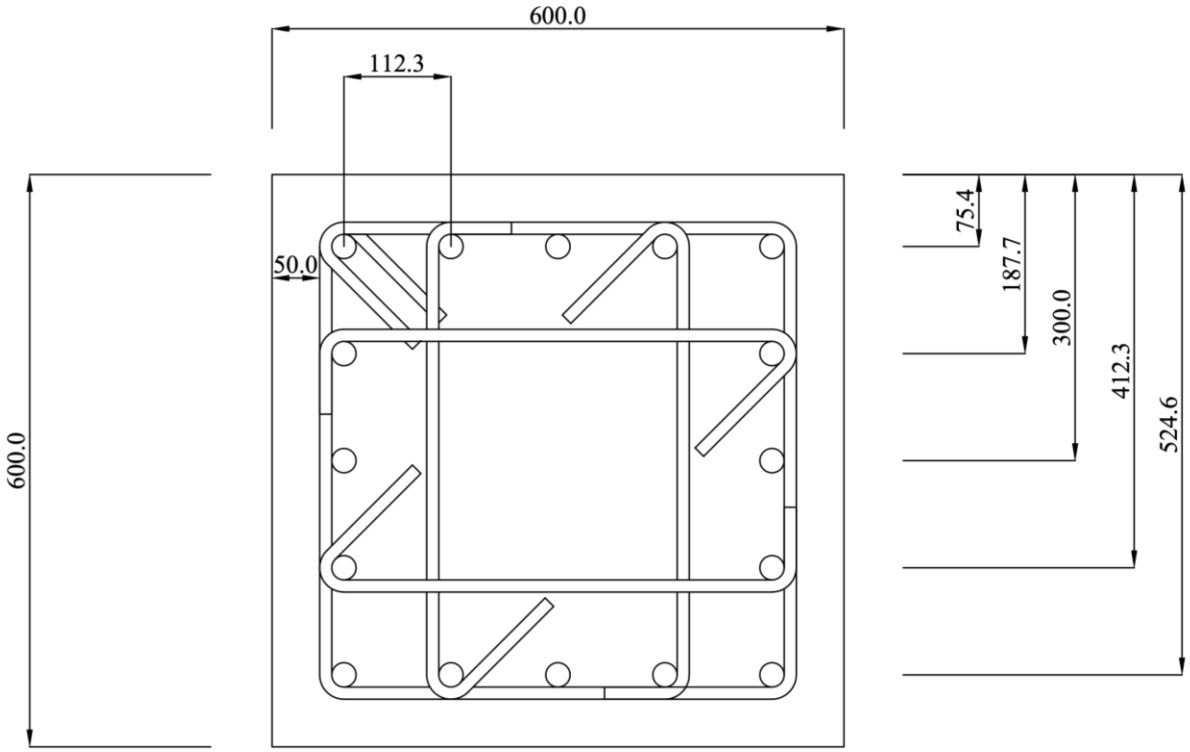
Moment-Curvature Behavior

Yu-Hung, Lin	N66144220
Ping-Xu, Lin	N66141125
Rhaditya Viansyah A.	E64148458

Table of content

1. Section properties
2. Parameter selection
3. OpenSees analysis
4. Xtract analysis
5. Comparison & Justification
6. Conclusions

Section properties



unit: mm

Longitudinal bars: #8-D25

Reinforcing ratio (longi.): 2.25%

Transverse bars: #4-D19

Section Area: 60 x 60 cm

Parameter selection

Concrete strength (MPa)	Steel grade (MPa)	
28	420	550
35		
42		

4 combinations were discussed in this report.

	1	2
1	case1: (28, 420),	case2: (28, 550)
	case3: (35, 420),	case4: (35, 550)
2	case5: (42, 420),	case6: (42, 550)

OpenSees analysis

OpenSees analysis

Material properties

Concrete (28、35、42MPa)

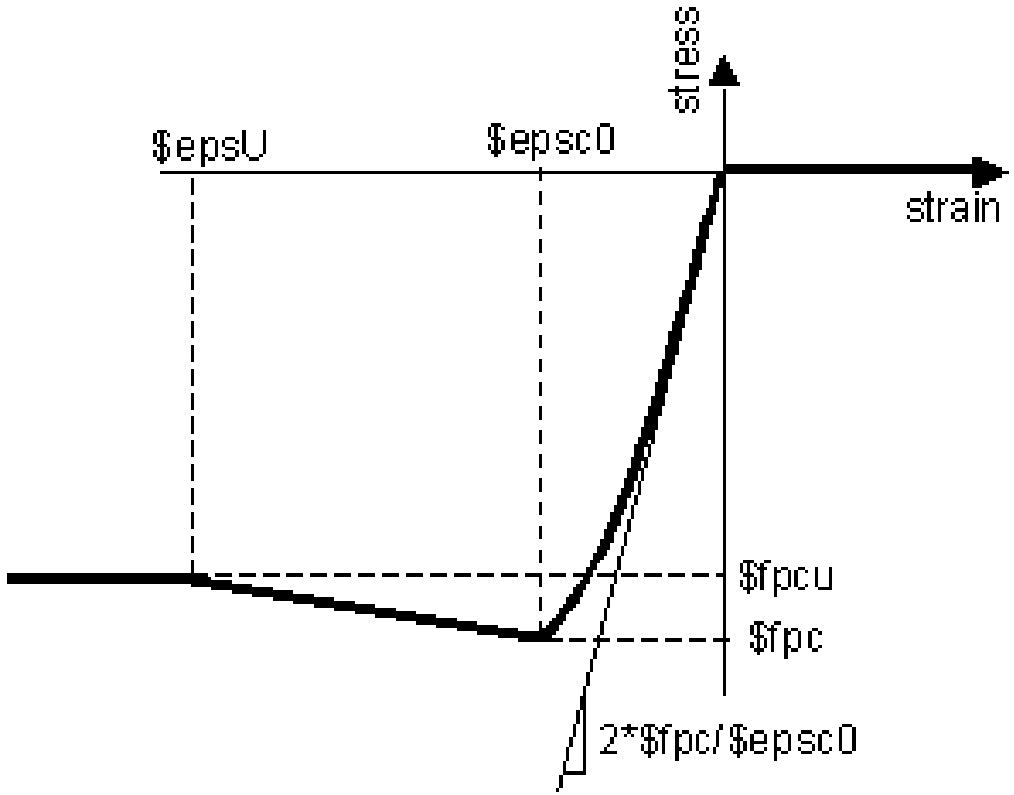
$f_c' = 28$、35、42 MPa

$\varepsilon_{c0} = 0.002$

$f_{pcu} = 0$ MPa

$\varepsilon_{cu} = 0.004$

Type: *Concrete01*



Reference: OpenSees Wiki

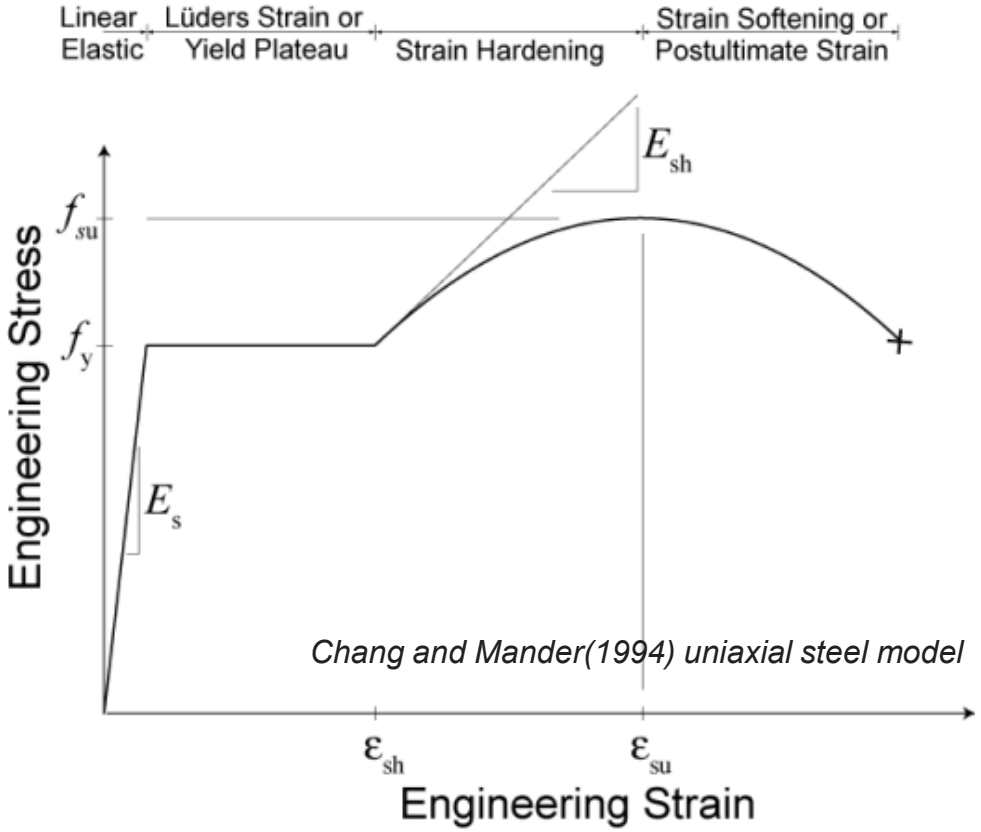
OpenSees analysis

Material properties

Reinforcement (420、550MPa)

$f_y = 420$、550MPa

Type: *Reinforcingsteel*



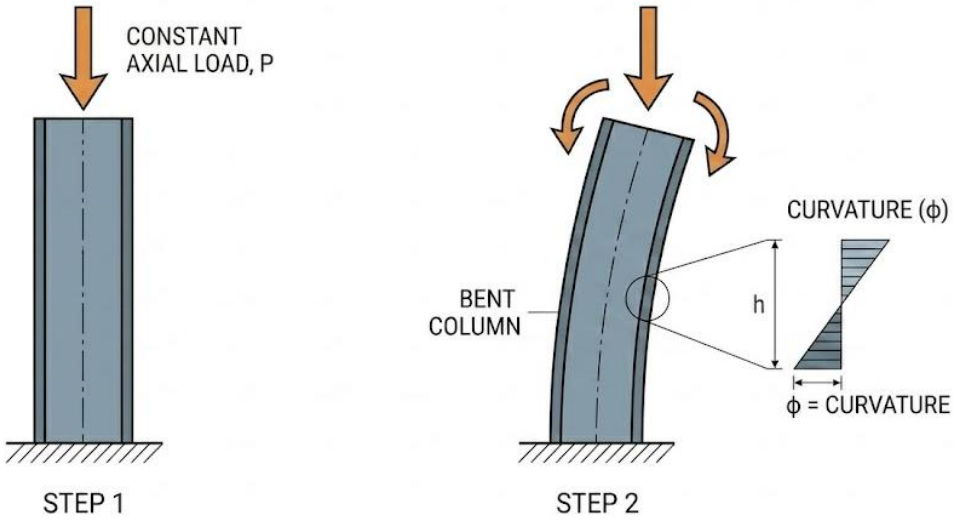
Reference: OpenSees Wiki

OpenSees analysis

Loading pattern

Phase 1: Apply a constant axial load (*20% axial load ratio*)

Phase 2: Increment the curvature (*Displacement control, DOF 3*)



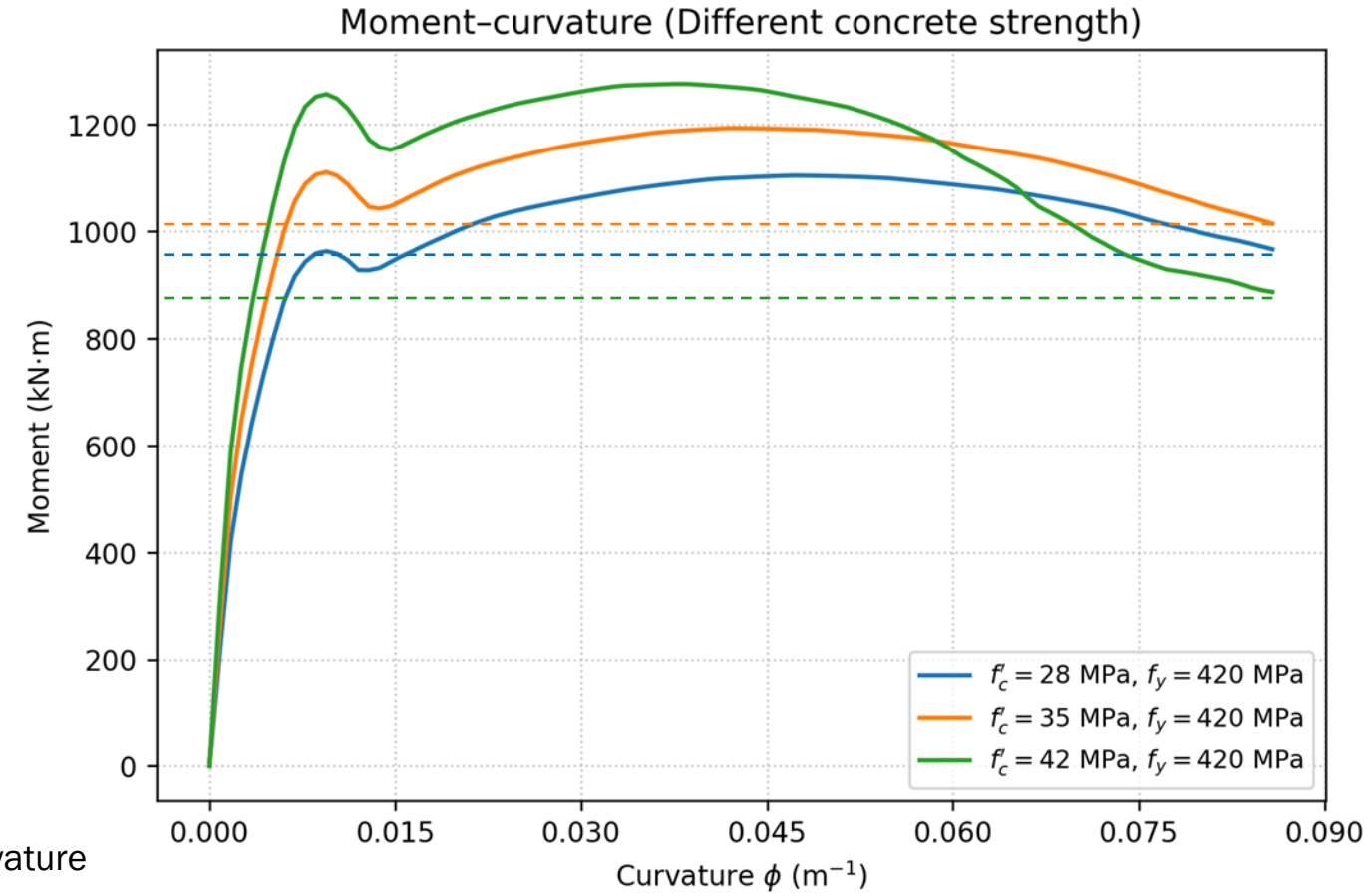
OpenSees analysis

Results

Different concrete strength

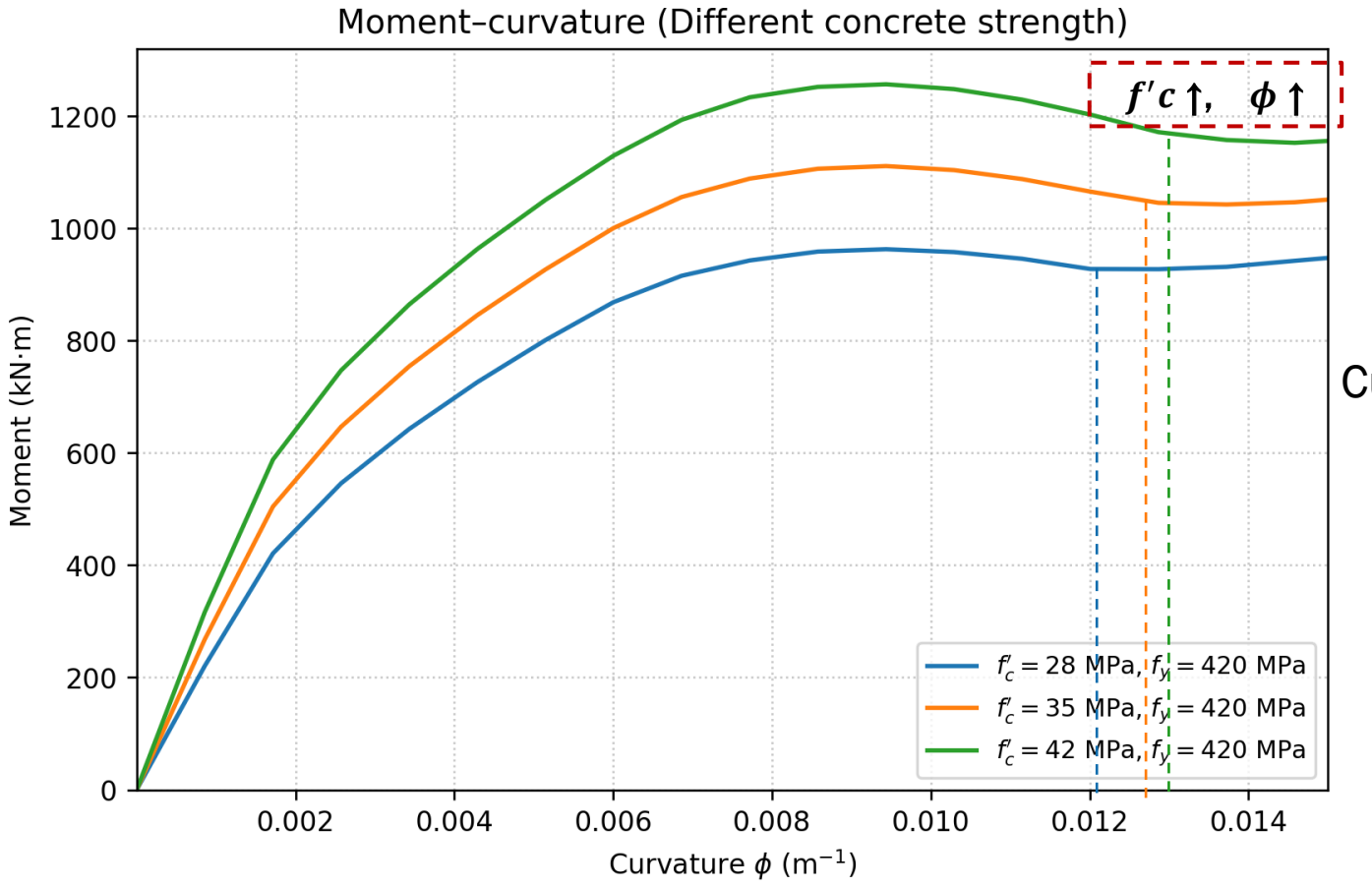
W / SD420

	M_u (kN-m)
case1	966.35
case3	1014.49
case5	886.82



$*M_{max}$ is the moment corresponding to the maximum curvature

OpenSees analysis



Curvature is equal to strain divided by neutral axis depth:

$$\phi = \frac{\varepsilon}{c}$$

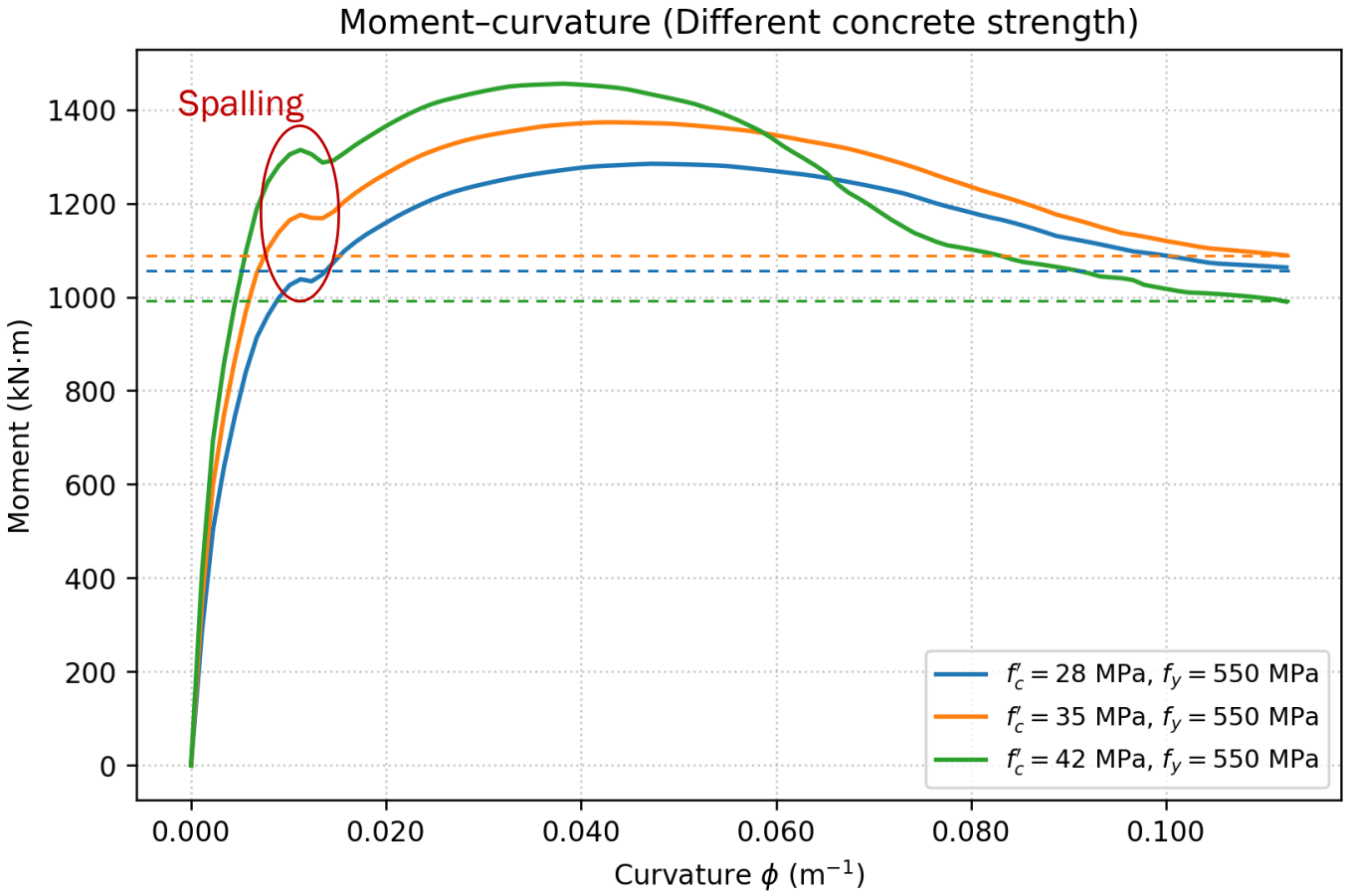
OpenSees analysis

Results

Different concrete strength

W / SD550

	M_u (kN-m)
case2	1063.07
case4	1088.82
case6	989.92

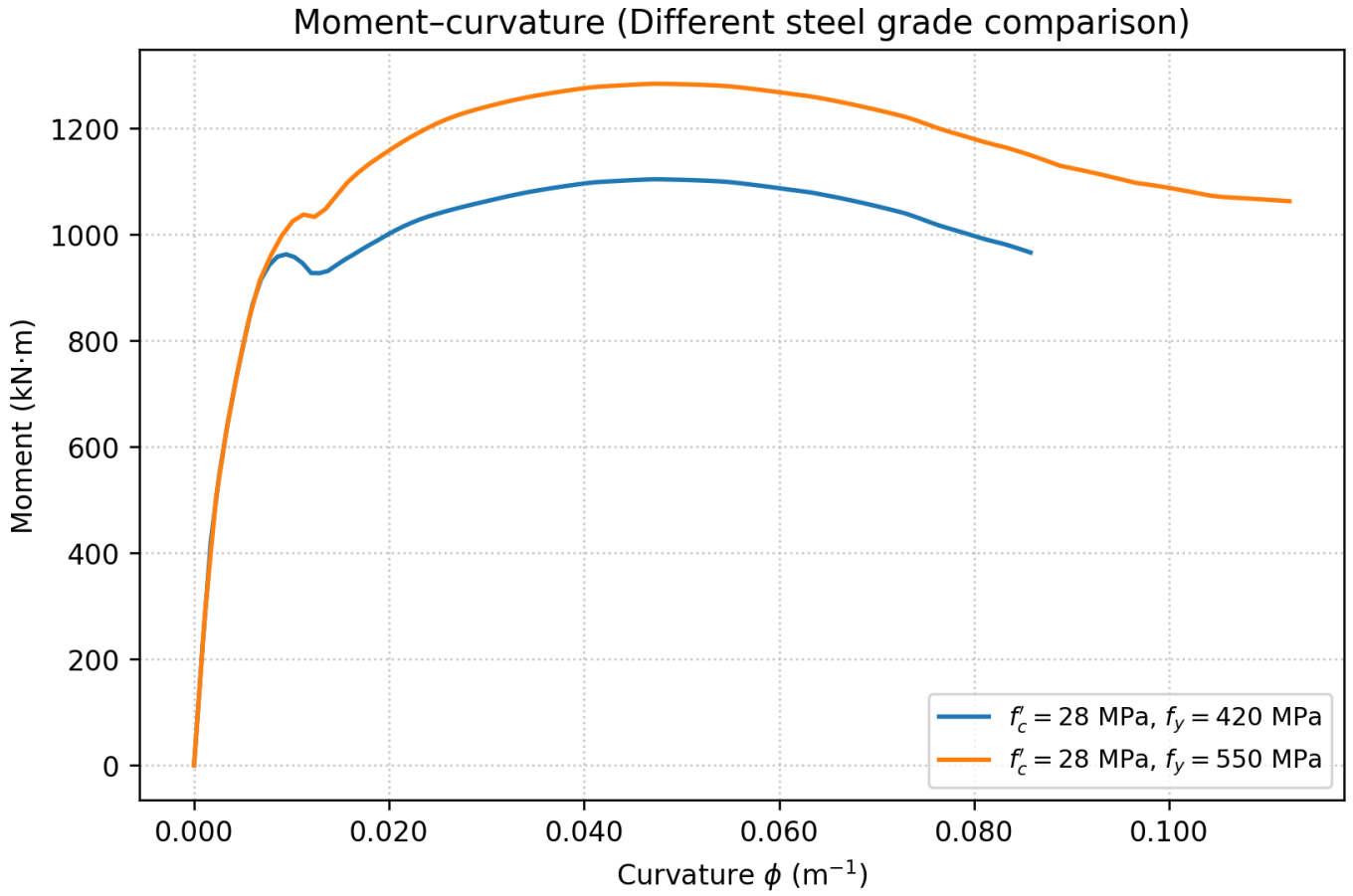


OpenSees analysis

Results

Different steel grade

W / 28MPa concrete

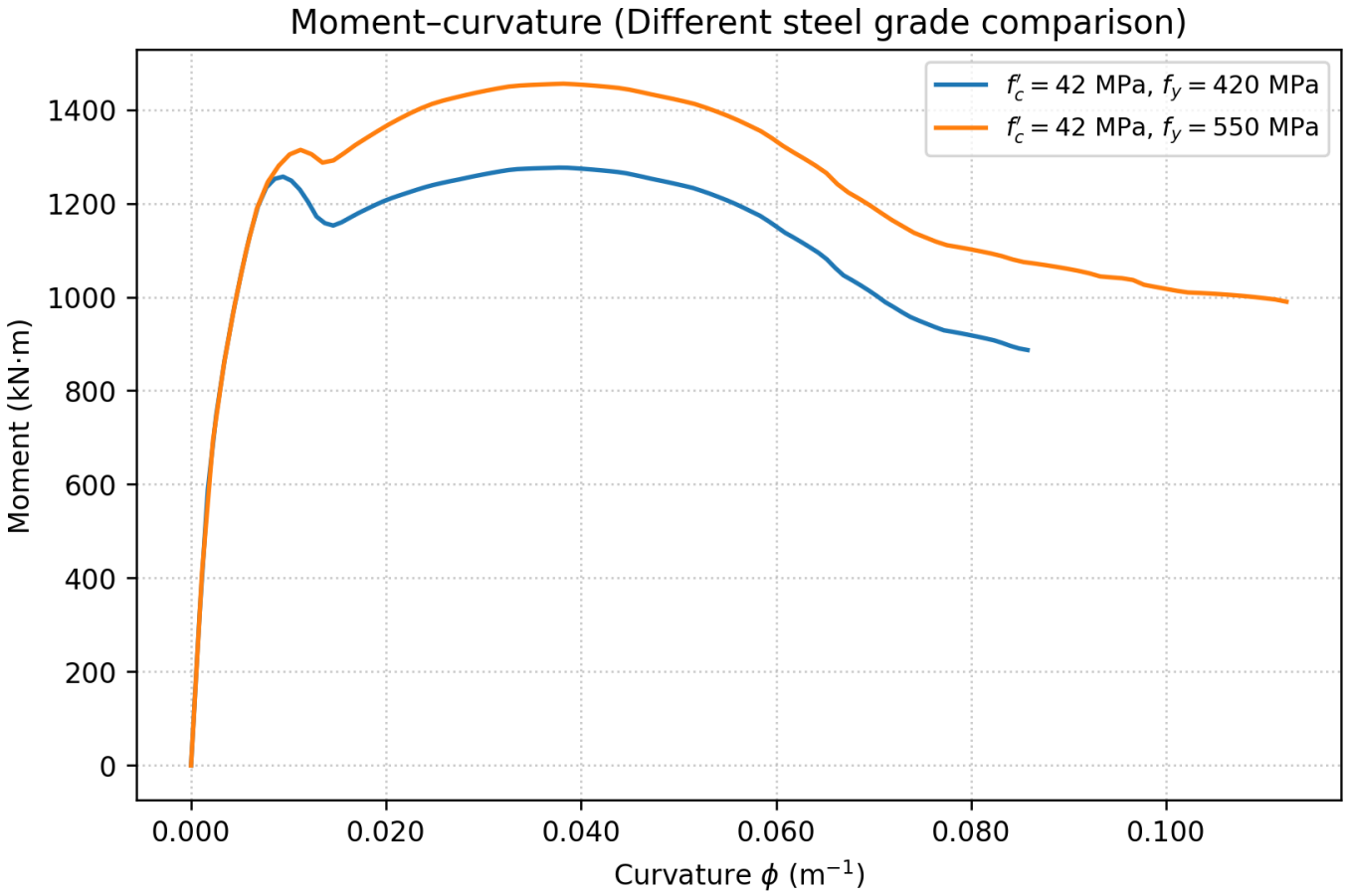


OpenSees analysis

Results

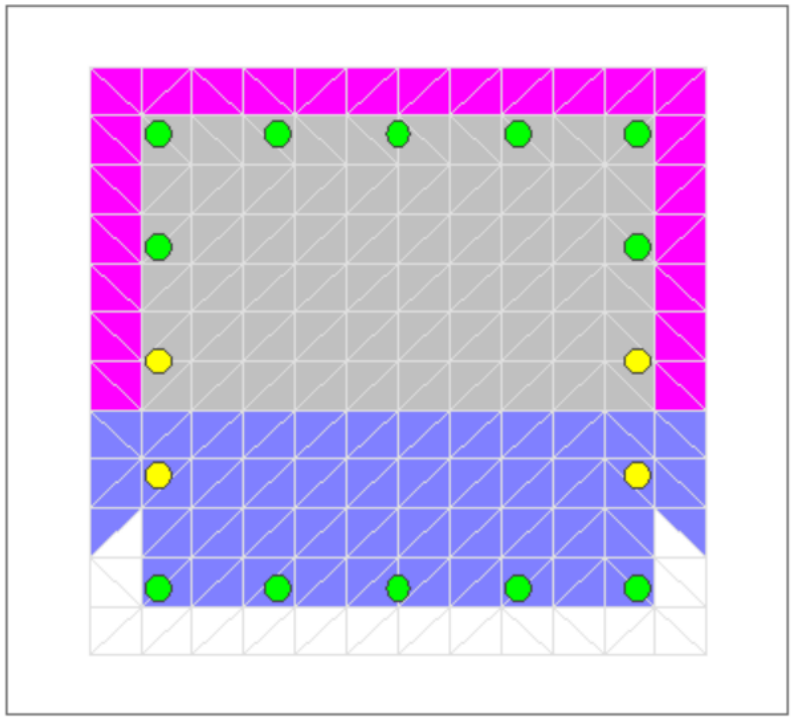
Different steel grade

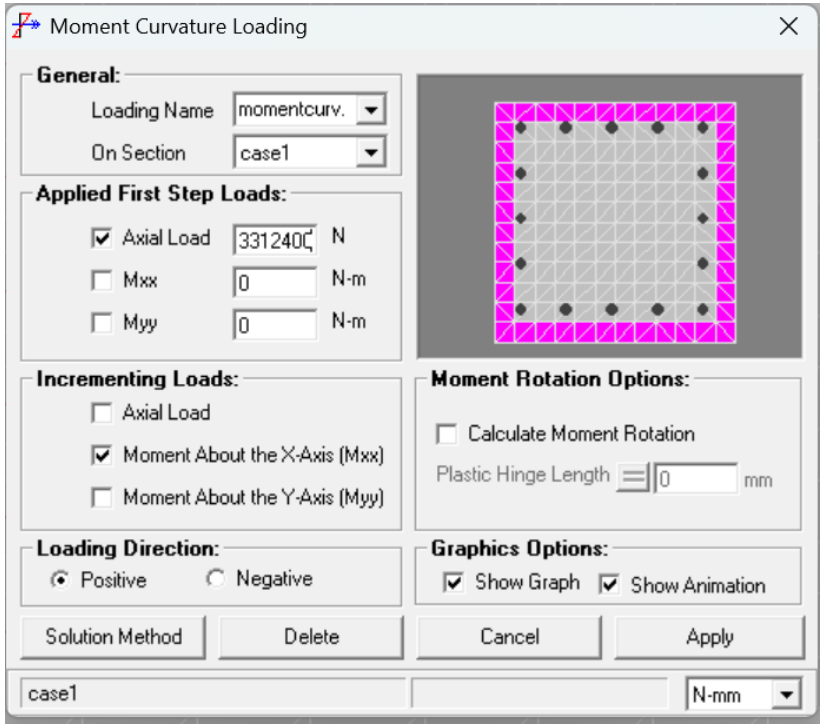
W / 42MPa concrete



Xtract analysis

Xtract analysis



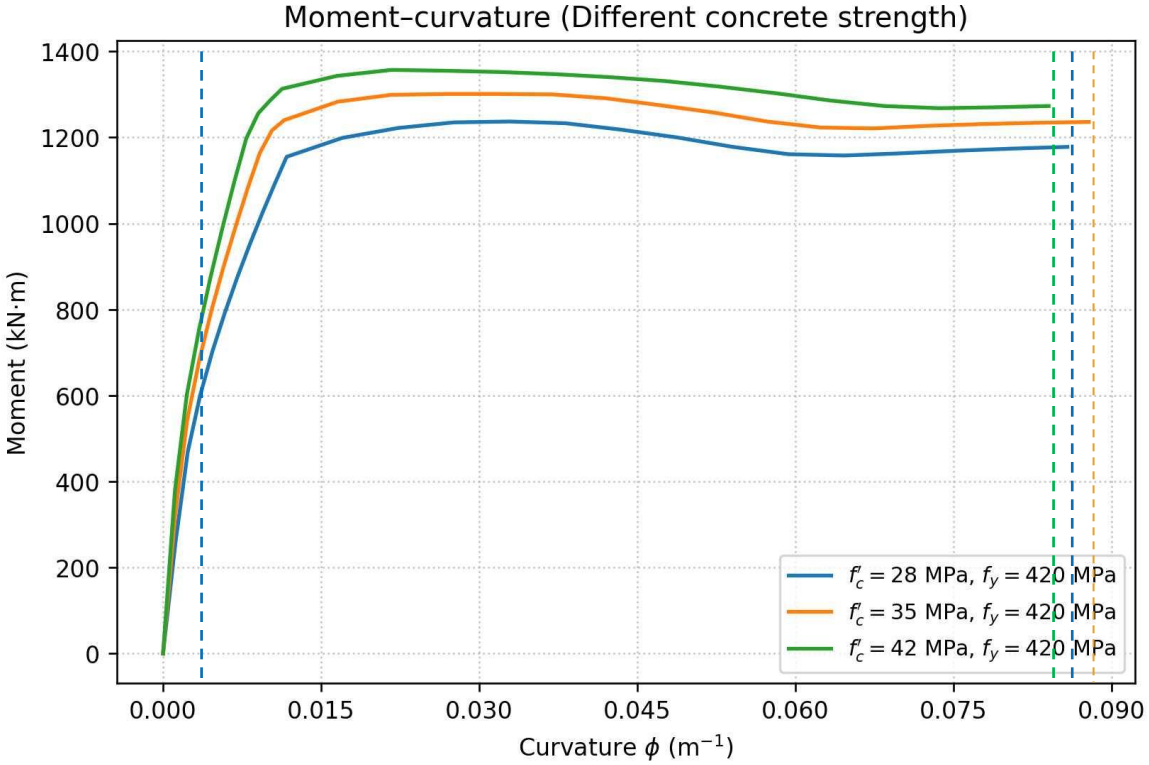


Color Codes

- White = Crushing of the concrete
- Blue = Compression in concrete
- Yellow = Yielding of rebar
- Green = Strai hardening of rebar
- Fuchsia and grey = Zero or tensile strain

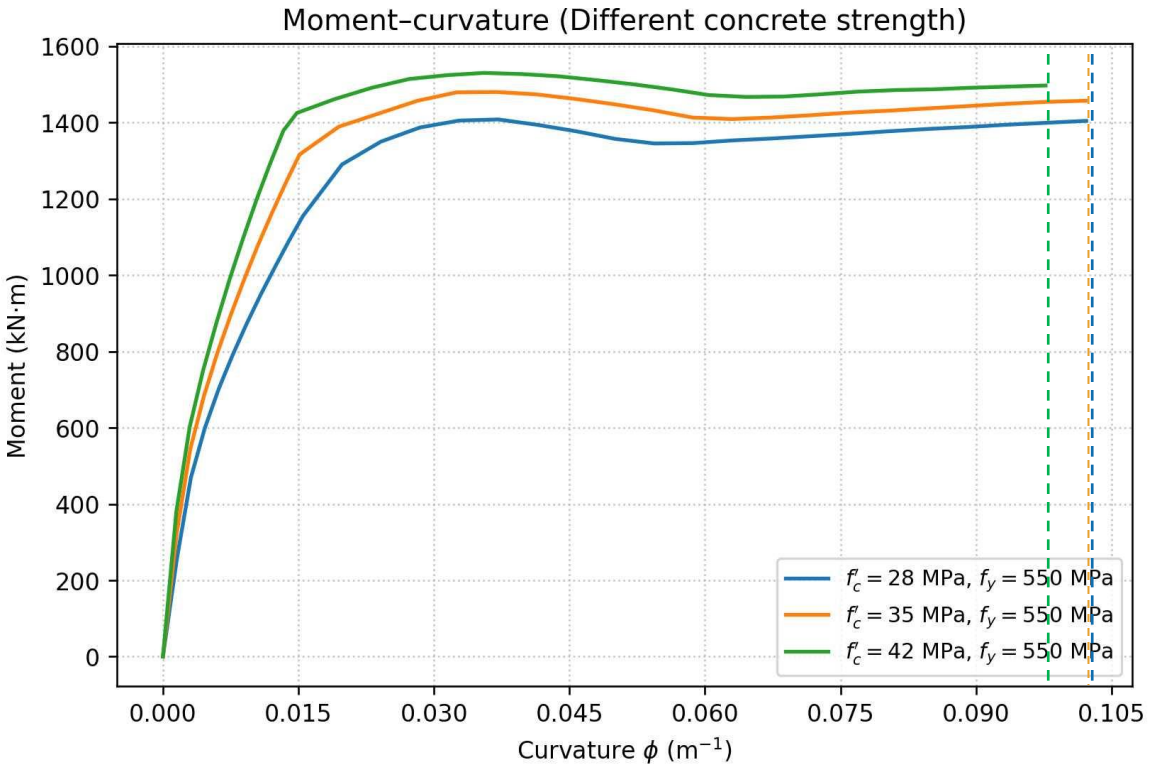
Xtract analysis

Concrete strength at fy 420



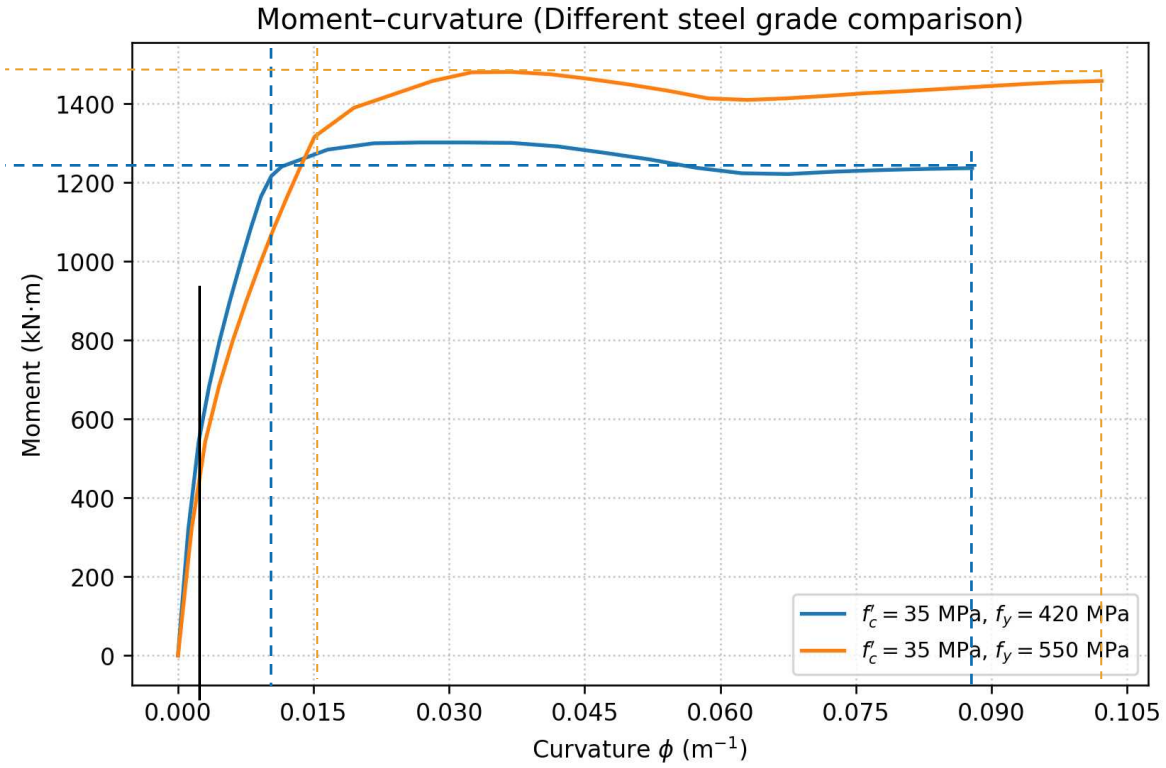
Xtract analysis

Steel grade 550



Xtract analysis

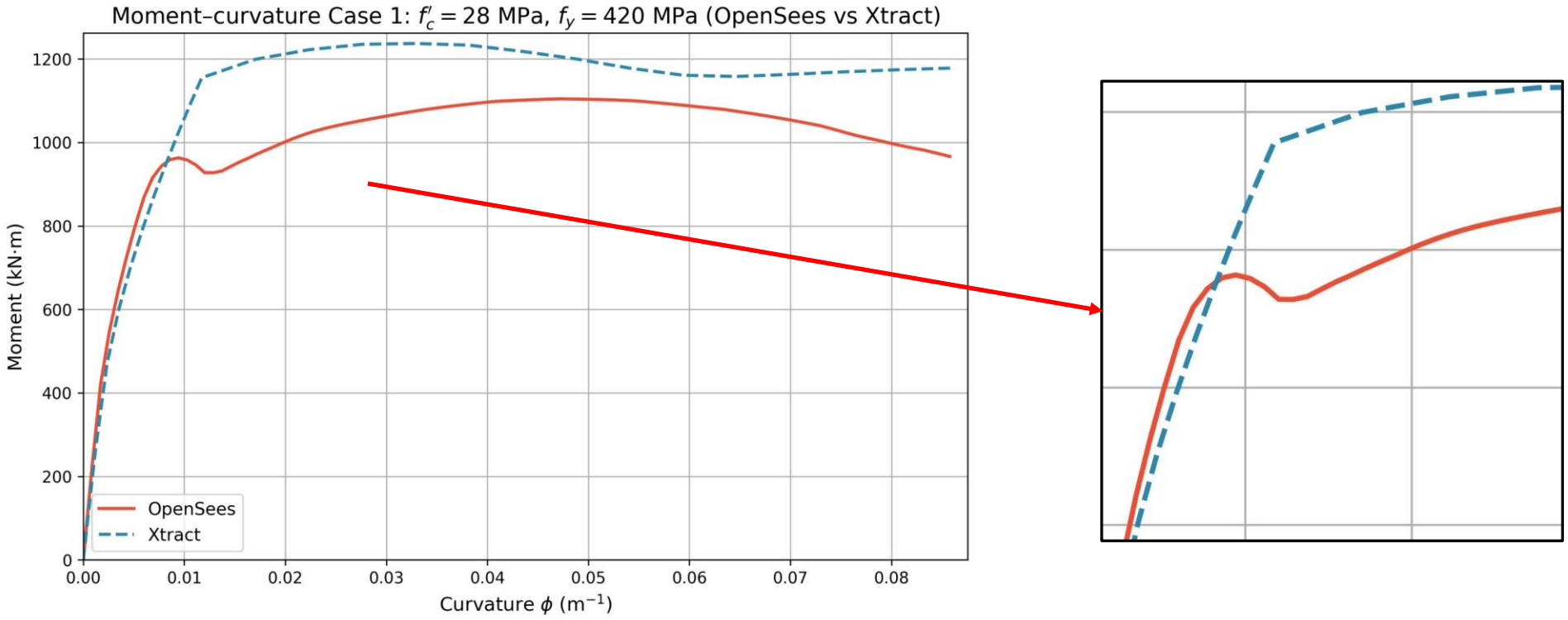
1,460 kN·m
vs.
1,280 kN·m



Concrete strength in different steel grade

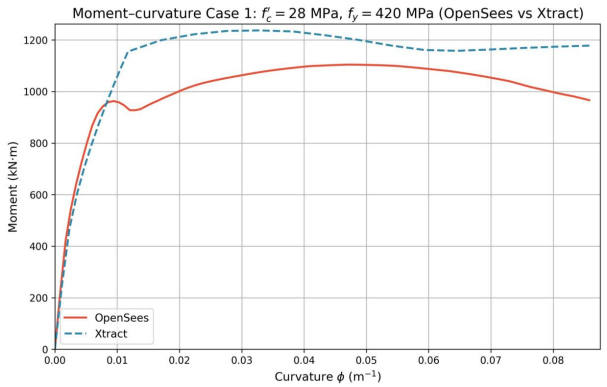
Comparison & Justification

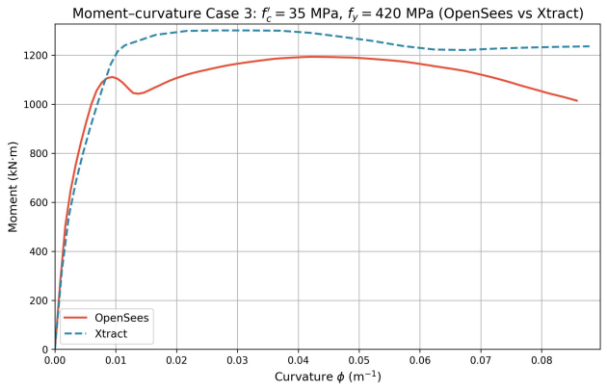
Results verification

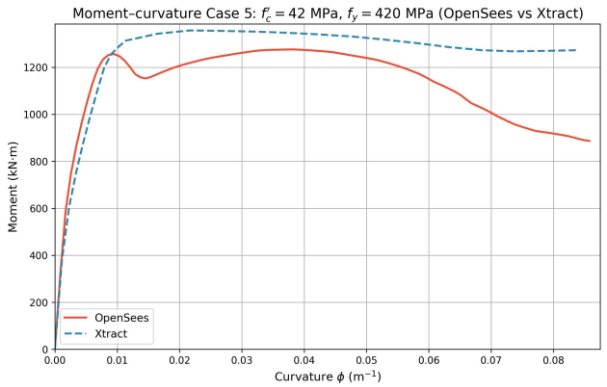


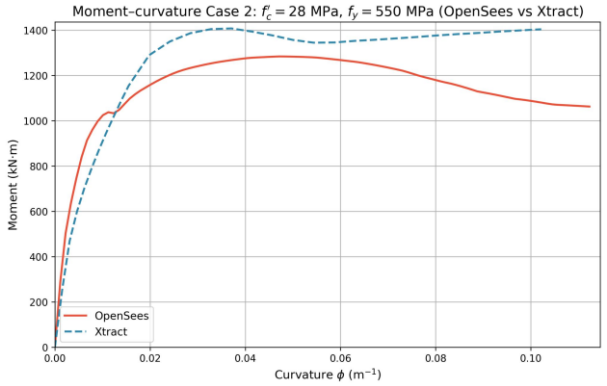
Case 1

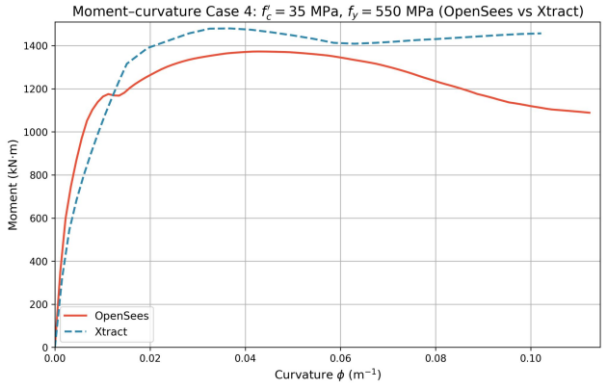
Results verification

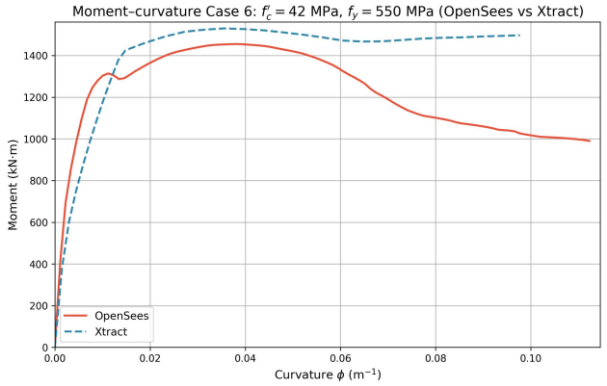












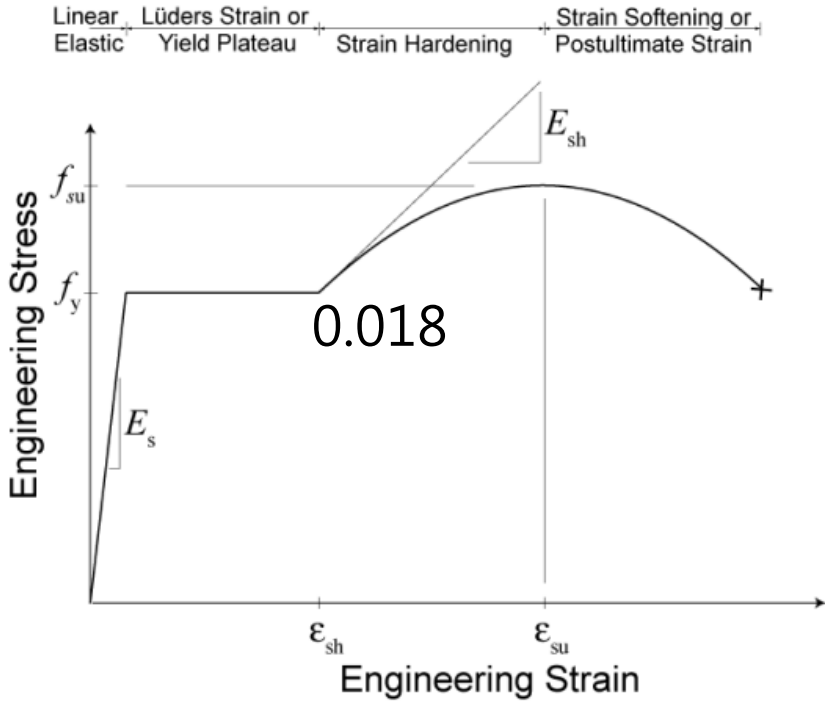
Results verification

Parameters unification

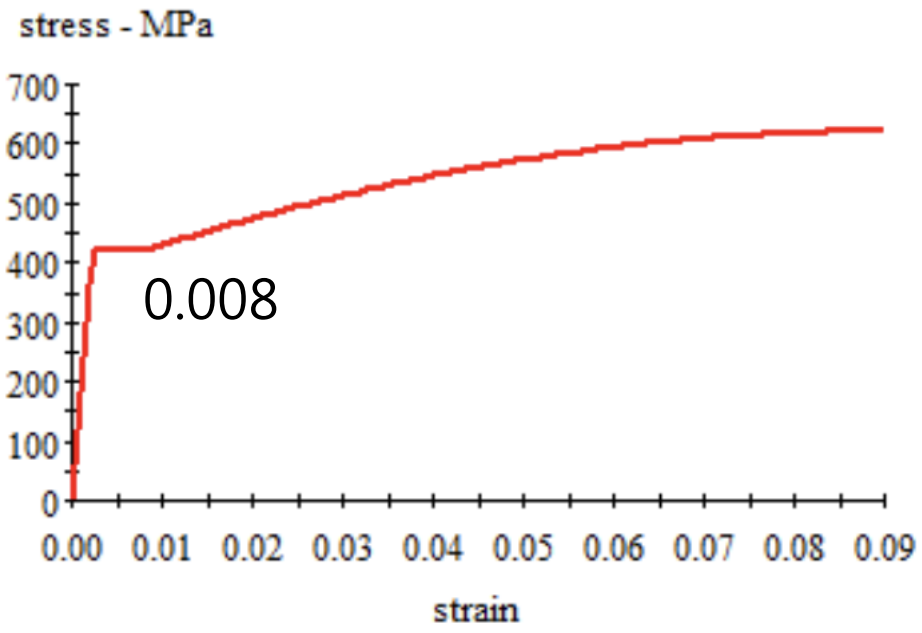
Steel

Concrete

Results verification

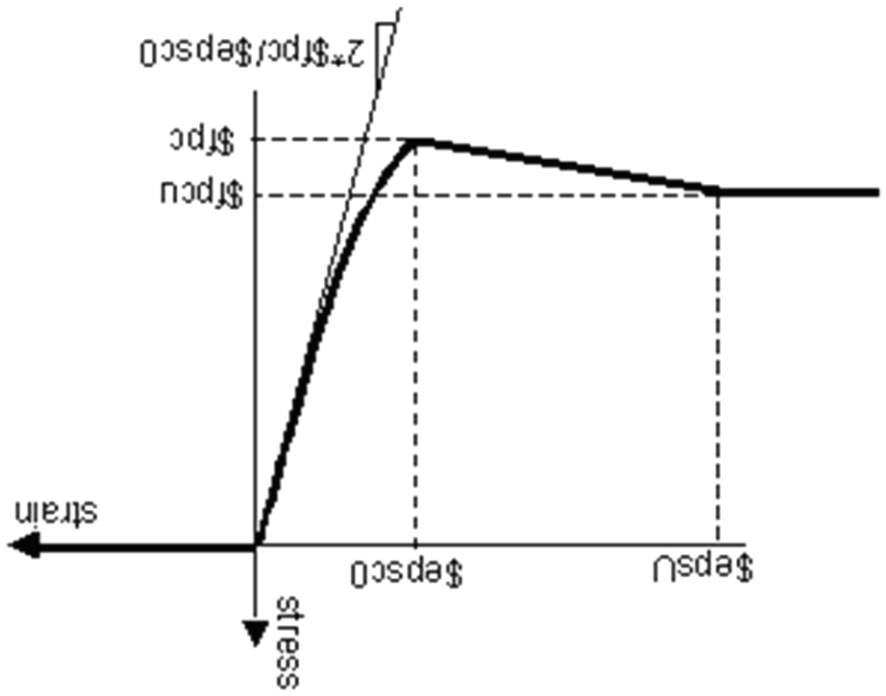


OpenSees: ReinforcingSteel

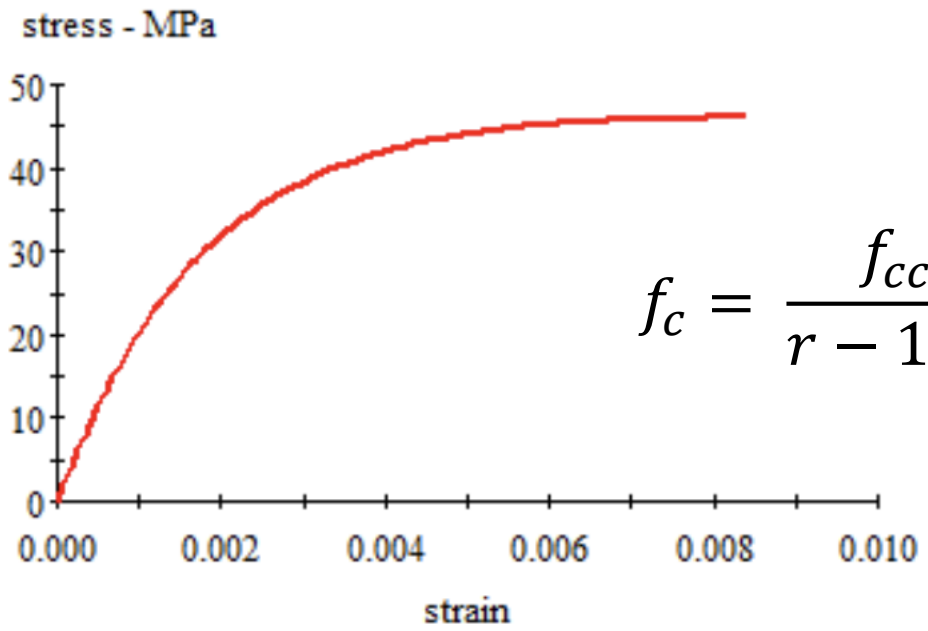


Xtract: Bilinear with Strain Hardening

Results verification



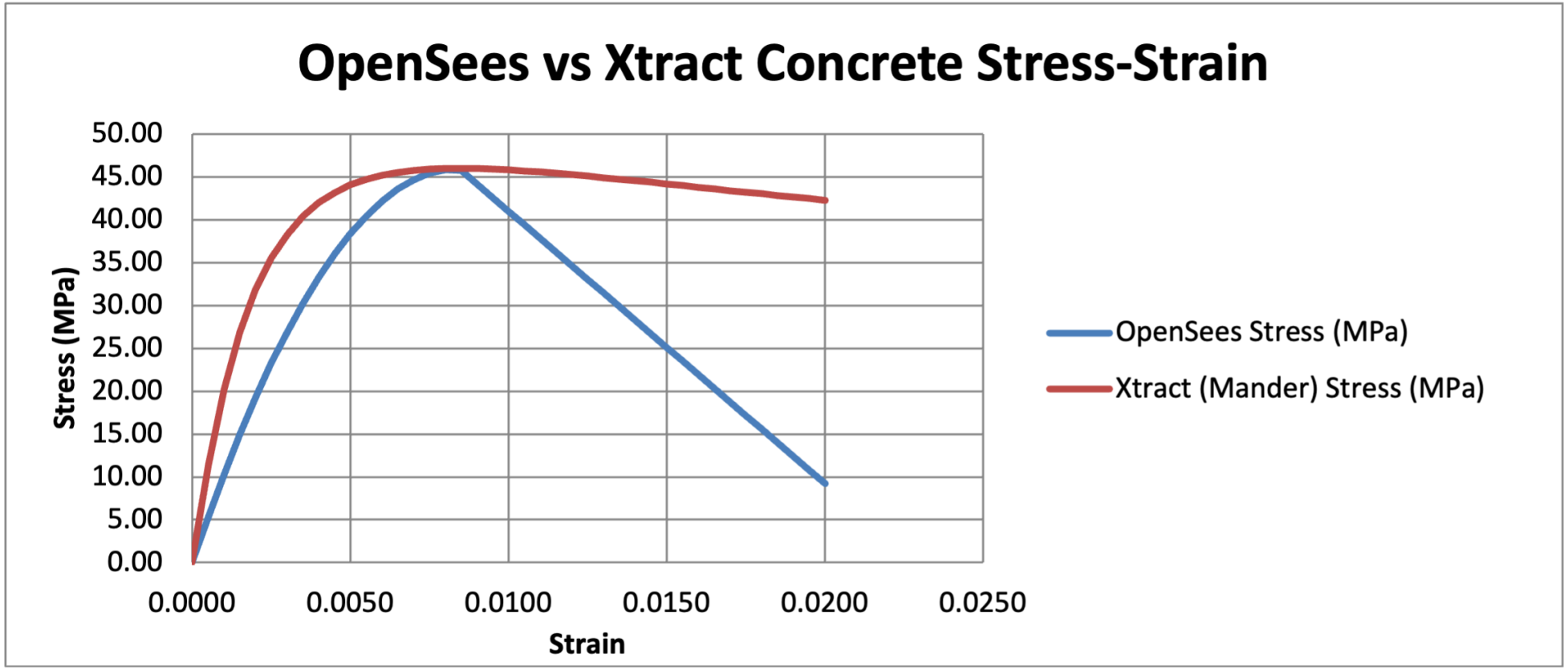
OpenSees: Concrete01



$$f_c = \frac{f_{cc} x r}{r - 1 + x^r}$$

Xtract: Mander Confined Concrete

Results verification



Conclusion

- Stronger materials can lead to higher moment.
- In the OpenSees model, the phenomenon of concrete cover spalling is clearly observable. However, this effect becomes less pronounced as f'c or fy increases.
- A significant portion of the discrepancies between the two models can be attributed to differences in their underlying material definitions and constitutive models.